Please upload your final project notebook here.

- The final project notebook should be commented, edited and properly ordered code you used to produce what you found (including visualizations), as well as markdown cells that describe your findings and allow someone reading the notebook to understand them, and whatever data files are necessary to run the notebook.
- Importantly, the final project notebook should **not** be a mere merging of EDA + modeling notebooks without any changes. Rather, it should be a unified and finalized notebook, and should be **substantively updated** to ensure **narrative consistency** and **remove any unnecessary code or discussions**.
- Notebooks with little or no revisions **will be penalized**. Thus, the final notebook should be **polished, revised, and finalized version** of your work.
- If it is impossible to have a single notebook (due to python version/dependency issues - for instance `pycaret` works only in python 3.9), you may upload 2 or more notebooks.
- You should put the Juptyer notebook(s) and data file(s) into a single folder and .zip them for submission.
- **Ideally, your notebook(s) should be capable of being run in a single go.**
- You will be graded on the criteria in the attached rubric. Note that both your **code** and the **narrative discussions** (question under investigation, why the question is important, the steps you took to answer the question, and the limitations and conclusions you were able to draw from the data) should be in the Python notebook.

Criteria	Ratings	Pts
Question, Hypothesis, Motivation clear and detailed statement of question under investigation and why the question is important		15 pts
Coding quality Code runs, is understandable, is efficient, is not superfluous and is revised, is adequately commented so that observer knows what is happening		20 pts
Visualization & tables Visualizations & tables are clear, are labeled appropriately, provide reader with understanding of data & insight into investigation of question		30 pts
Narrative and Methodology Coherent textual narrative description (in as many places as necessary) of how you went about answering question, including finding and assembling data, eda, modeling, validation, etc. where appropriate; quality of written expression is important--clarity, organization, use of English, with reference to course readings and/or outside readings. Incorporation of feedback from prior assignments.		50 pts
Interpretation of the results Detailed interpretation of the restuls, whereby the results are carefully interpreted, with clear explanations of: what the findings show, how they relate to the original question or hypothesis, the significance and implications of the results and how this fits with the class readings/outside readings		35 pts
Conclusions, limitations and future work detailed well-written narrative about what you are able to conclude about your question after analyzing the data, including possible gaps and how another investigator might fill those; discussion of how well your methods served your goals and what you might do differently; discuss legal, policy, and ethical implications as well as future areas of research - with reference to course readings and/or outside readings.		25 pts
Overall clarity of writing, style and explanation + citations and references how well your textual narratives are written. clearness and accessibility of these methods and models is what one should aim for - in addition to citing appropriate literature that seems related and, if you used Generative AI for coding, appropriate citation of its use		25 pts
Bonus points going beyond and above what is expected in terms of originality and inventiveness in finding answer to your question either with data sources or analysis methods--worth up to 10 pts extra credit		0 pts

For the web presentation you will hand in a **Markdown notebook** (consisting of markdown cells with formatting information, which you must save as a .md file) along with everything (including the images of data visualizations) that you want to present, all in a folder and zipped up for handing in here. **You must submit them as a .zip file.**

In order to save the jupyter notebook (.ipynb) as a markdown file (.md), you would need to use nbconvert as follows:

1. run in your notebook:
   !jupyter nbconvert --to markdown --output-dir="your_dir" your_notebook.ipynb
2. To produce a clean presentation without code cells, you should use:
   !jupyter nbconvert --to markdown --TemplateExporter.exclude_input=True --output-dir="your_dir" your_notebook.ipynb
3. After the conversion:
   - Ensure that all generated files (including images) are included
   - Place them in a single folder
   - Compress the folder into a `.zip` file for submission

For more details on nbconvert syntax and markdown see:

- https://reproducible-science-curriculum.github.io/publication-RR-Jupyter/02-exporting_the_notebook/index.html
- https://nbconvert.readthedocs.io/en/latest/index.html

Make sure your notebooks are in a presentable state by previewing them in VSCode. The markdown presentation should essentially be a copy of the final project notebook, and thus be clear and coherent, with parts and subsections that make sense.

You may put your name in the notebook or not, as you see fit, although if there is no name in the notebook you will not be able to claim credit publicly if/when these notebooks are published or you want to post them.

The rubric below describes how the web presentation will be evaluated (note that even though engagement is mentioned in the "clarity" criterion, engagement is a holistic sort of thing).

Rubric

Criteria	Ratings	Pts
Formatting, Submission, and Technical Execution - Notebook is correctly exported from .ipynb to .md using nbconvert - Code cells are appropriately excluded (i.e., presentation-focused) - All required assets (e.g., images, figures) are included and correctly displayed - Files are organized in a single folder and submitted as a .zip file - No broken links, missing files, or rendering issues		10 pts
Structure and Organization - Presentation is clearly structured with logical sections and subsections - Eeasy to follow from introduction to conclusion - Use of headings, spacing, and formatting enhances readability - Content is organized for a web audience, not just copied from the notebook		10 pts
Clarity of Communication - Writing is clear, concise, and well-edited - Technical ideas are explained in an accessible way - Terminology is used appropriately and consistently - The presentation is understandable to a reader without prior exposure to the project		10 pts
Visualizations and Supporting Material - Visualizations are well-integrated into the narrative - Figures are clearly labeled and easy to interpret		10 pts
Overall coherence - Overall coherence and of the notebook - Narrative feels intentional and polished		10 pts
		Total points: 50